Business Forecasting Capstone Project

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2026-03-15

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
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## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
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```

```
## Loading required package: gsubfn
```

```
## Loading required package: proto
```

```
## Warning in fun(libname, pkgname): couldn't connect to display ":0"
```

```
## Loading required package: RSQLite
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```
## Skipping install of 'hrbrthemes' from a github remote, the SHA1 (d3fd0294) has not ch
anged since last install.
##   Use `force = TRUE` to force installation
```

```
## Rows: 9800 Columns: 18
## ── Column specification ────────────────────────────────────────────────────────────
──────
## Delimiter: ","
## chr (15): Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer ...
## dbl  (3): Row ID, Postal Code, Sales
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 48 Columns: 2
## ── Column specification ────────────────────────────────────────────────────────────
──────
## Delimiter: ","
## dbl  (1): CSI
## date (1): Date
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

1. Introduction

a. Business Interest and Objective

This superstore sells many different categories of consumer and commercial products, ranging from home goods, to office supplies, electronics, and other consumer products. Recently the company has began an initiative to study sales in each category in order to identify which categories are the top performers during different times of the year. This information will allow the store to anticipate upcoming demand, and shift their promotion focus to these top performers ahead of time in order to maximize sales in these main categories while minimizing expenses in underperforming categories. By maximizing sales in popular categories, the store will anticipate demand and generate more income for future initiatives.

b. The Problem to Solve

By analyzing 4 years worth of sales data, we will identify trends, seasonality, and outliers in key categories, and determine which will likely have the highest customer demand in the upcoming month. A series of regression models will be used to determine Adjusted-Rsq values to explain the effect of predictor values on each sales category. These models will take into account past lagged values as well as the University of Michigan Consumer Sentiment Index values to forecast the next month's sales in each main category.

2. Data Description and Exploration

```
# show the total number of records in the dataset using nrow()
total_orders <- nrow(SalesData)
# show the first 6 rows of the dataset using head()
head(SalesData)
```

Row ID <dbl>	Order ID <chr>	Order Date <chr>	Ship Date <chr>	Ship Mode <chr>	Customer ID <chr>	Customer Name <chr>
1	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute
2	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute
3	CA-2017-138688	12/06/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff
4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell
5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell
6	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman

6 rows | 1-8 of 18 columns

```
# Do some data wrangling to prepare the data for this project

monthly_totals <- as.data.frame(
  table(format(as.Date(SalesData$`Order Date`, "%m/%d/%Y"), "%Y-%m"))
)

total_months <- nrow(monthly_totals)

States=length(unique(SalesData$State))
Categories=length(unique(SalesData$Category))
SubCategories=length(unique(SalesData$`Sub-Category`))

# Overview of dataset
total_orders
```

```
## [1] 9800
```

```
total_months
```

```
## [1] 48
```

```
States
```

```
## [1] 49
```

```
Categories
```

```
## [1] 3
```

```
SubCategories
```

```
## [1] 17
```

```
# Format Consumer Sentiment Index by Year/Month
names(CSI) <- c("Date", "CSI")
CSI$Date <- format(as.Date(CSI$Date), "%Y-%m")  # <-- was "%Y-%m-%d"

# Aggregate sales by month and format dates by Year/Month
total_monthly_sales <- aggregate(
  Sales ~ format(as.Date(`Order Date`, "%m/%d/%Y"), "%Y-%m"),
  data = SalesData,
  sum
)

names(total_monthly_sales)[1] <- "Date"
names(total_monthly_sales)[2] <- "Total Sales"
total_monthly_sales
```

Date <chr>	**Total Sales** <dbl>
2015-01	19546.163
2015-02	11678.994
2015-03	6716.044
2015-04	12455.482
2015-05	15165.051
2015-06	11884.169
2015-07	10075.740
2015-08	26797.763
2015-09	17158.932

Date	Total Sales
<chr>	<dbl>
2015-10	10112.641

1-10 of 48 rows Previous **1** 2 3 4 5 Next

```
# Create dataframe aggregating monthly sales by category
category_monthly <- aggregate(
  Sales ~ format(as.Date(`Order Date`, "%m/%d/%Y"), "%Y-%m") + Category,
  data = SalesData,
  sum
)

# Aggregate monthly sales by category
category_totals <- aggregate(Sales ~ Category, data = category_monthly, sum)
names(category_monthly)[1] <- "Date"
category_monthly
```

Date	Category	Sales
<chr>	<chr>	<dbl>
2015-01	Furniture	8507.469
2015-02	Furniture	1693.850
2015-03	Furniture	3574.659
2015-04	Furniture	1803.958
2015-05	Furniture	2798.821
2015-06	Furniture	5544.520
2015-07	Furniture	5157.028
2015-08	Furniture	8061.612
2015-09	Furniture	4496.787
2015-10	Furniture	3565.225

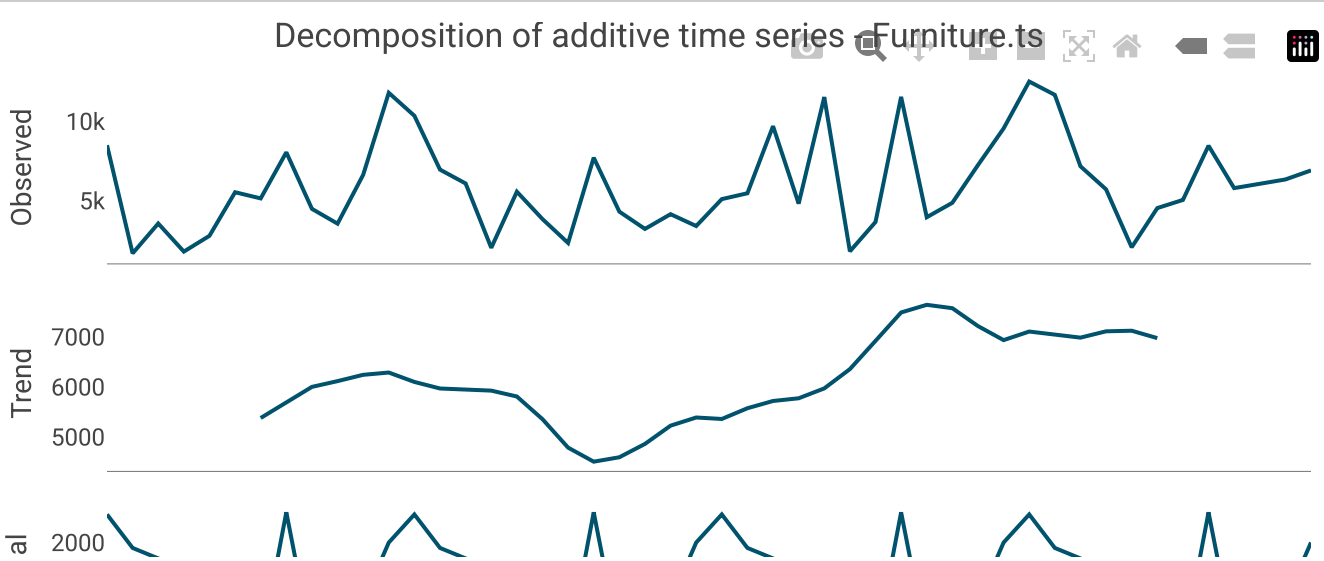
1-10 of 144 rows Previous **1** 2 3 4 5 6 … 15 Next

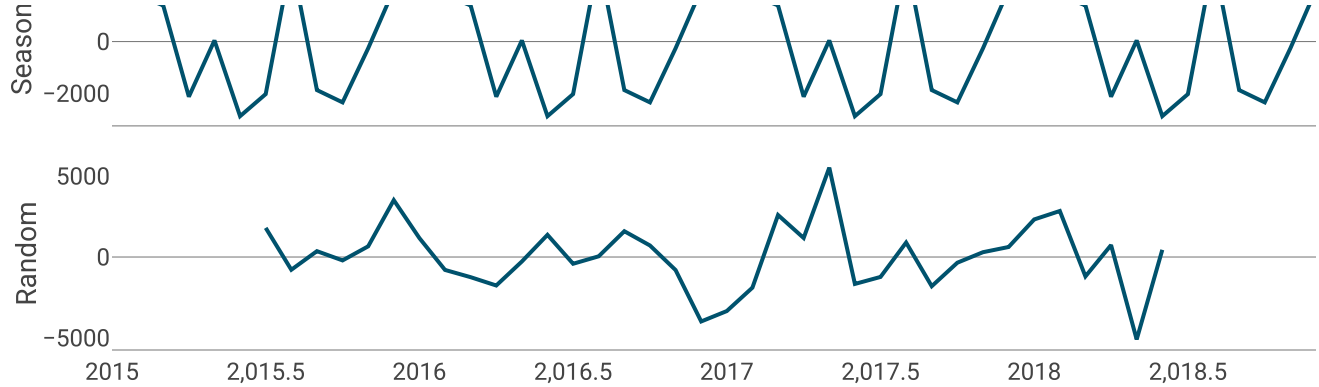
```
# Organize aggregated monthly sales by category into a table and add Consumer Sentiment
Index column
monthly_category_sales <- category_monthly %>%
  pivot_wider(names_from = Category, values_from = Sales, values_fill = 0)
  monthly_category_sales <- merge(monthly_category_sales, CSI, by = "Date", all.x = TRU
E)
   names(monthly_category_sales)[names(monthly_category_sales) == "Office Supplies"] <-
"Office"
monthly_category_sales
```

Date <chr>	Furniture <dbl>	Office <dbl>	Technology <dbl>	CSI <dbl>
2015-01	8507.469	4602.364	6436.330	98.1
2015-02	1693.850	4241.836	5743.308	95.4
2015-03	3574.659	1635.017	1506.368	93.0
2015-04	1803.958	4023.952	6627.572	95.9
2015-05	2798.821	10173.814	2192.416	90.7
2015-06	5544.520	3567.483	2772.166	96.1
2015-07	5157.028	2814.226	2104.486	93.1
2015-08	8061.612	4749.963	13986.188	91.9
2015-09	4496.787	5715.757	6946.388	87.2
2015-10	3565.225	2881.400	3666.016	90.0

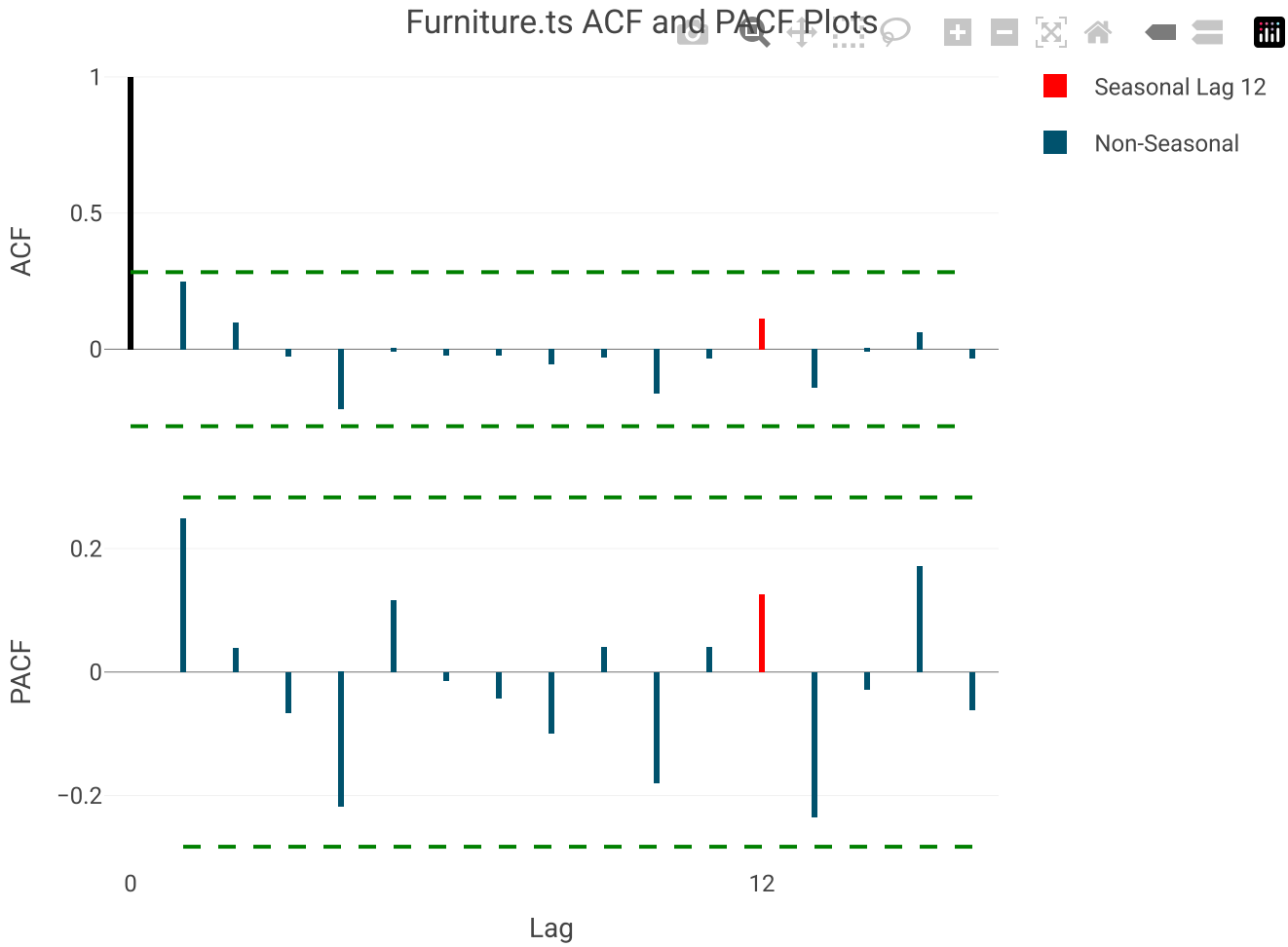
1-10 of 48 rows Previous **1** 2 3 4 5 Next

```
# Create time series for each category
Furniture.ts <- ts(monthly_category_sales$Furniture,
                start = c(2015, 1),
                frequency = 12) # Monthly data

Technology.ts <- ts(monthly_category_sales$Technology,
                start = c(2015, 1),
                frequency = 12)  # Monthly data

Office.ts <- ts(monthly_category_sales$Office,
                start = c(2015, 1),
                frequency = 12) # Monthly data

# Visualize time series graphs and ACF, PACF
ts_decompose(Furniture.ts)
```

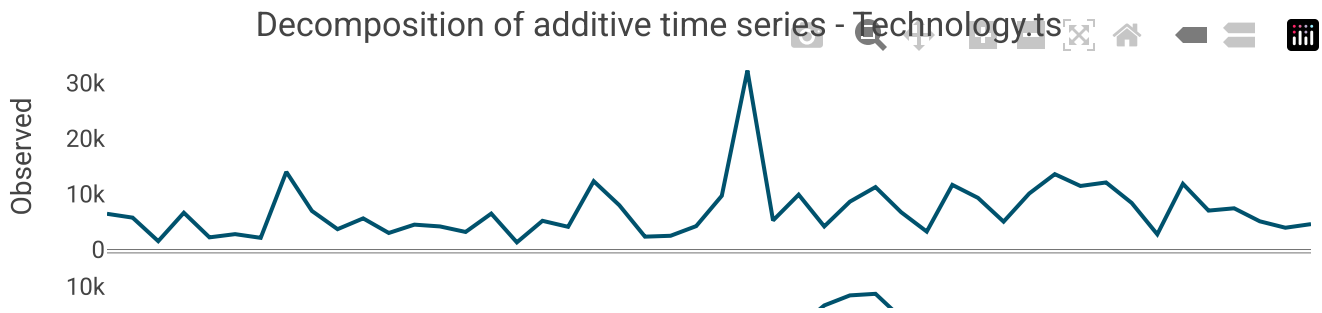


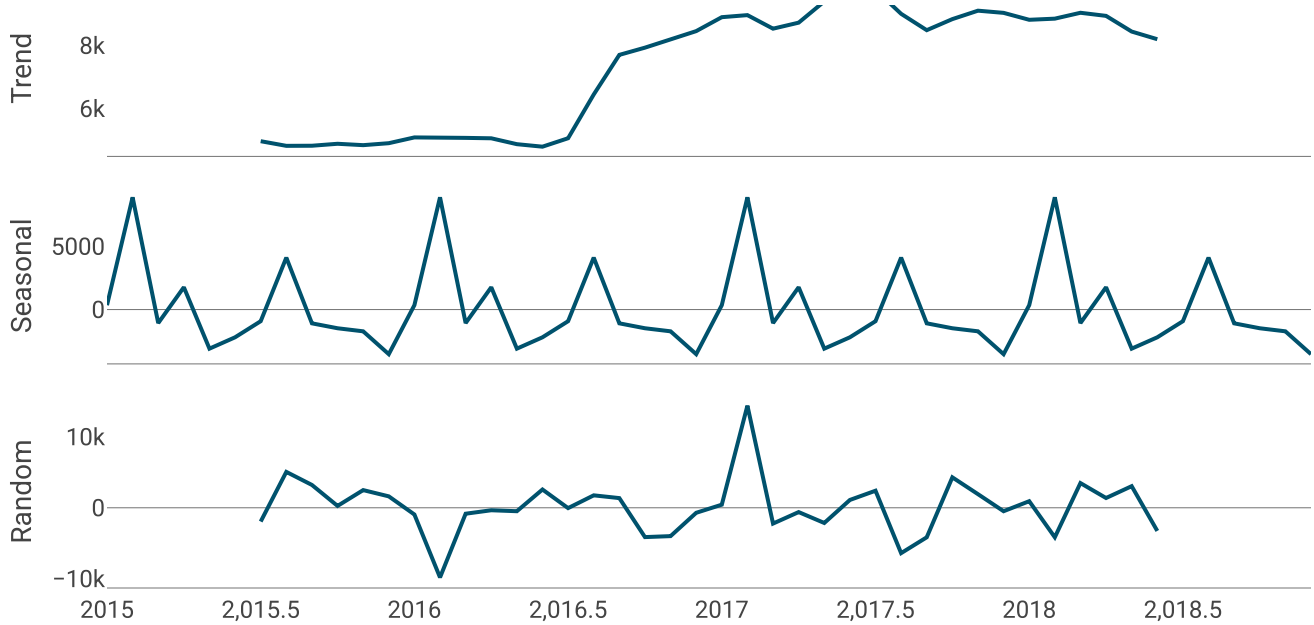



```
ts_cor(Furniture.ts)
```

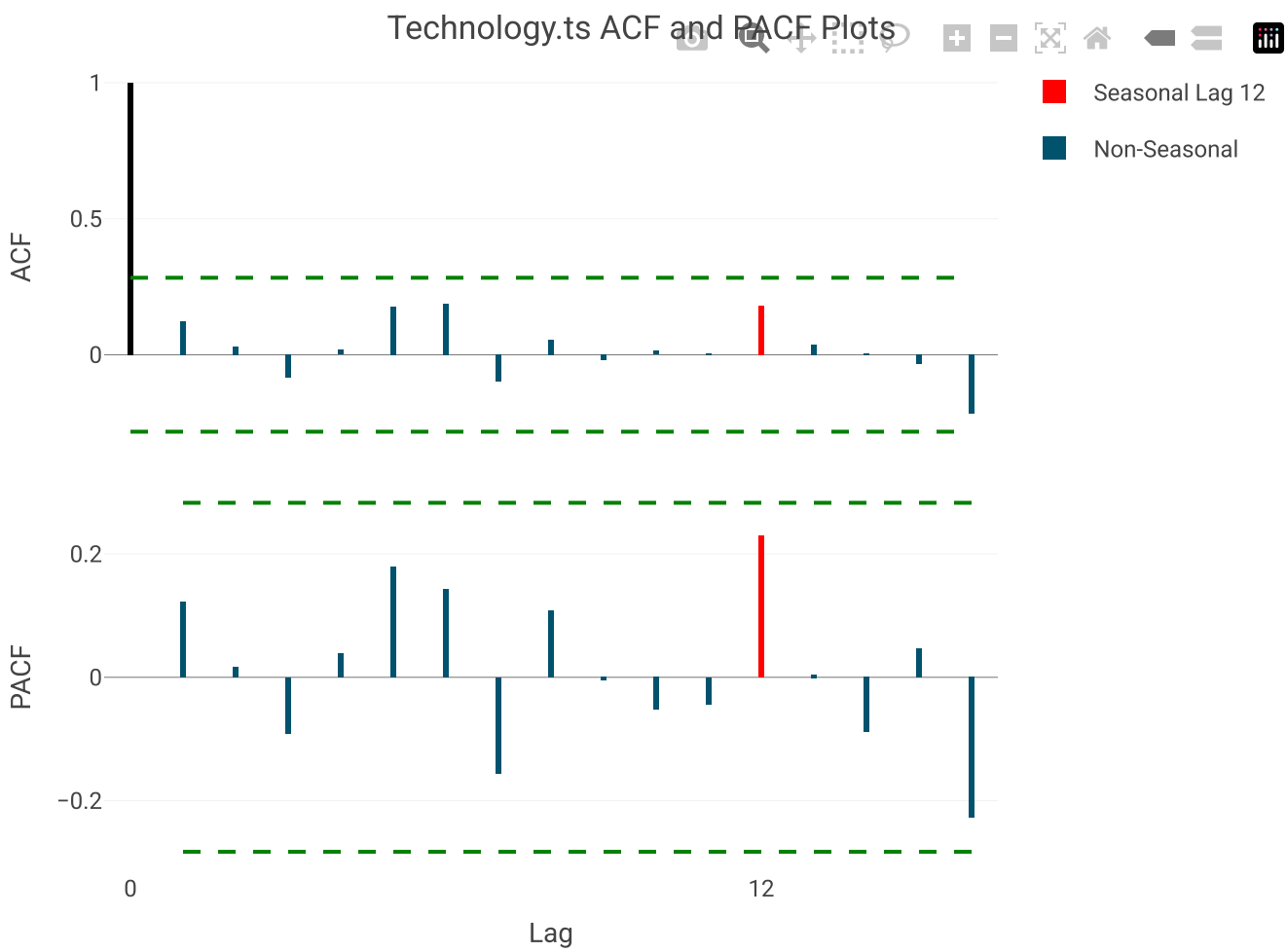



```
ts_decompose(Technology.ts)
```

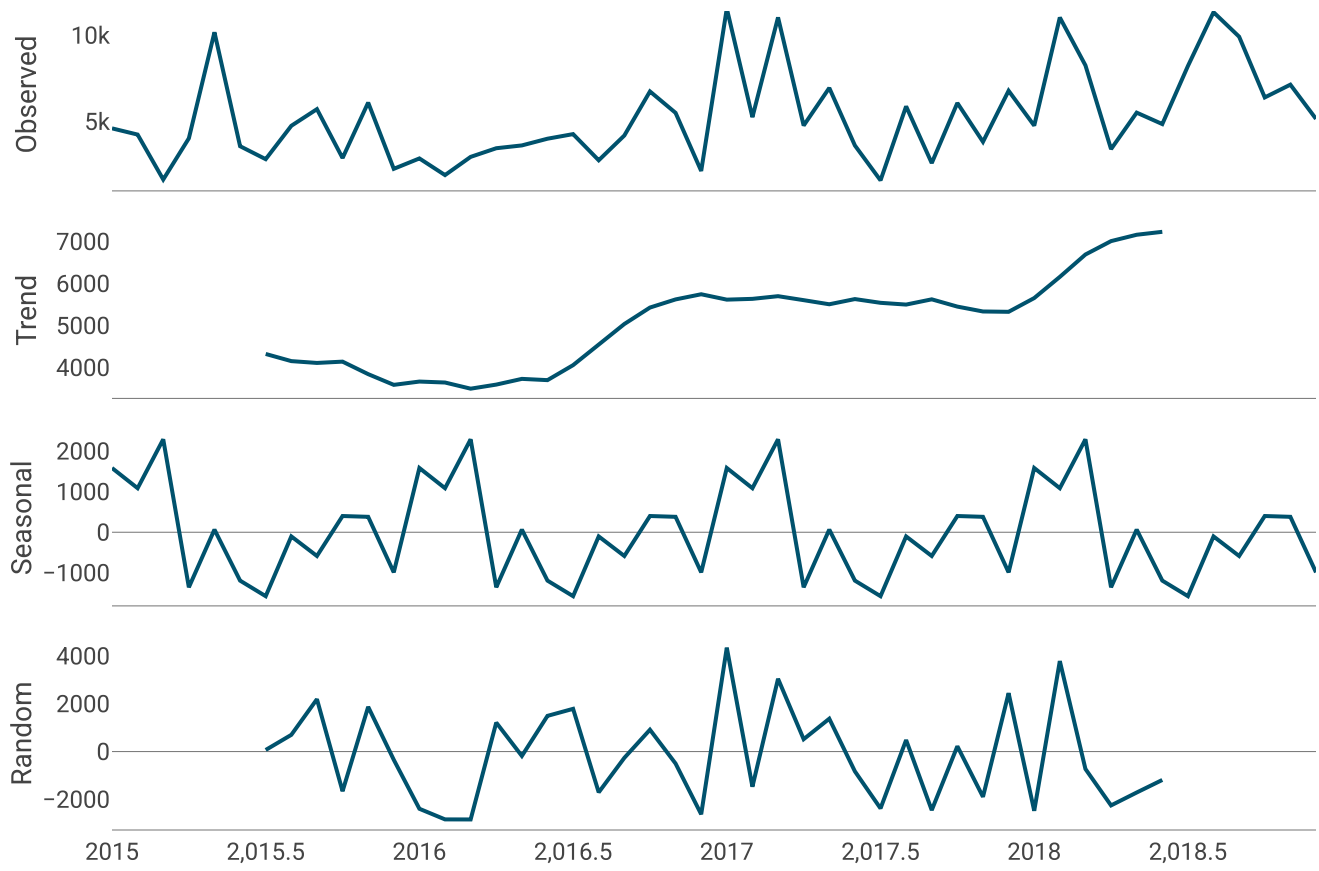




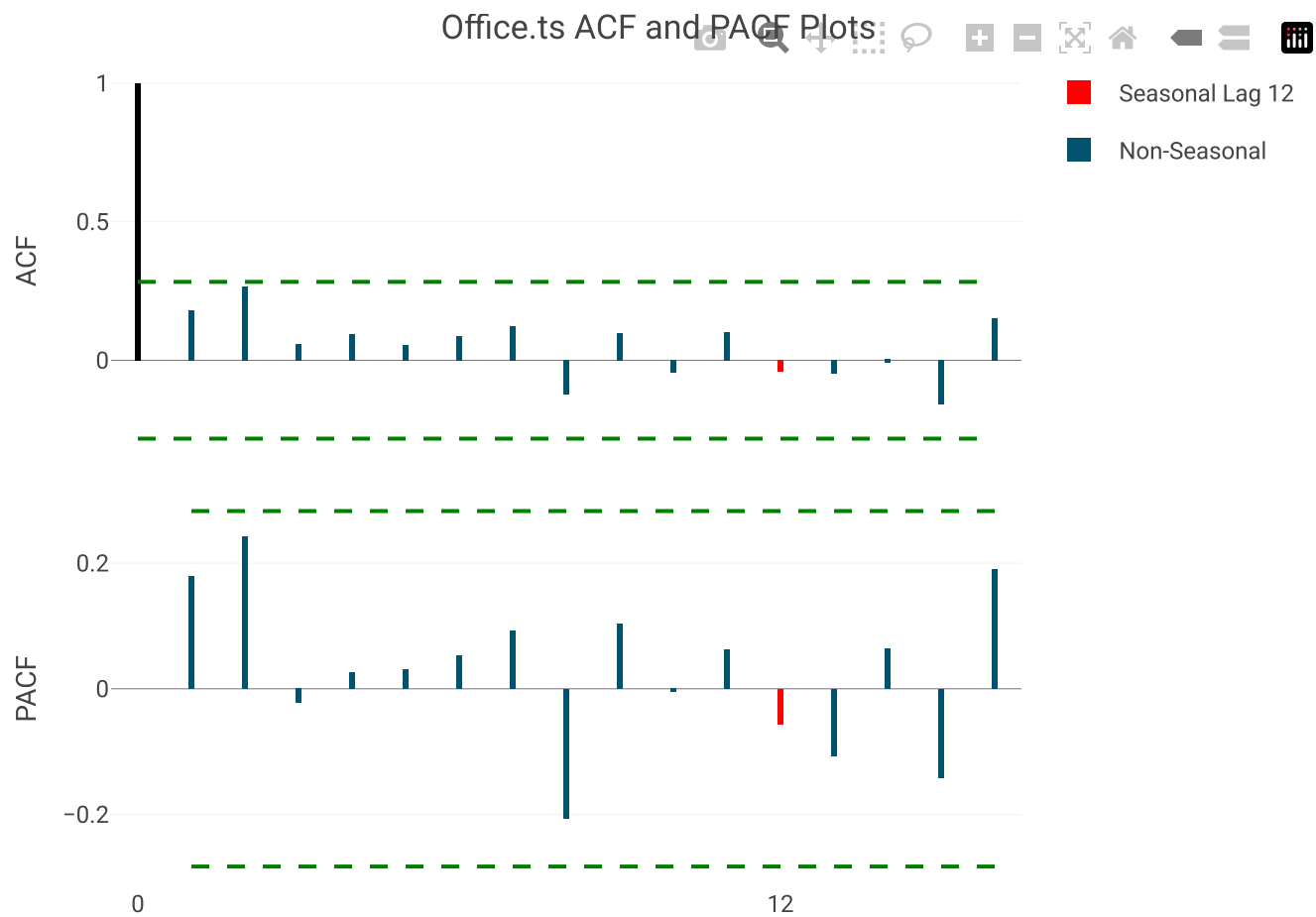

```
ts_cor(Technology.ts)
```




```
ts_decompose(Office.ts)
```




```
ts_cor(Office.ts)
```



Lag

DESCRIBE DATA COLLECTION & TRENDS,CORRELATIONS,SIMILARITIES

The above data is drawn from a collection of point-of-sale transactions at a national superstore chain that sells consumer products in multiple categories. These transactions record information such as Order Date, ZIP Code, City, State, Region, Product ID, Customer ID, Category, Sub-Category, and Product Name. Each entry is stored in a transactional database in which each row references the relevant sale information for each transaction. This dataset spans four years from Jan 2015 to Dec 2018, and totals 9800 entries from stores in 49 states, with products from 3 categories, and 17 sub-categories.

Upon visualizing these transactions we can see that each category fluctuates in sales throughout the year. These fluctuations show up in the seasonal visualization every few months, with the largest and longest increases occurring at the end of each year. There is also a strong upward trend over the entire dataset which also shows acceleration at each year-end. This could reasonably be attributed to increasing holiday shopping as the holiday season approaches. The largest apparent increase in sales occurred in the Technology category from December-January 2017, however when looking at the associated ACF and PACF charts for each category, there are no significant spike outside the normal distribution range, so this could be seen as normal seasonal activity within a normal distribution.

3. Models and Analysis

a. Forecasting models

Furniture sales trend projection

```
# Create working dataframe with Furniture, Total Sales, and CSI
total_Furniture <- sqldf('select Date, Furniture, `Total Sales`, CSI from monthly_catego
ry_sales
                          join total_monthly_sales using(Date)')
names(total_Furniture)[3] <- "TotalSales"

# Create 1-lagged Total Sales values for Furniture SLR
total_Furniture$TotalSaleslag1 <- lag(total_Furniture$TotalSales, 1)
total_Furniture.SLR <- na.omit(total_Furniture)

# Build Furniture SLR predicting Furniture sales using 1-lagged Total Sales as predictor
Furniture.SLR <- lm(Furniture~TotalSaleslag1, data=total_Furniture.SLR)

# Furniture Multiple Regression Model 1
# Add 2-lagged Total Sales and remove missing values
total_Furniture$TotalSaleslag2 <- lag(total_Furniture$TotalSales, 2)
total_Furniture.MLR1 <- na.omit(total_Furniture)

# Build Furniture MLR1 predicting Furniture sales using 1-lagged and 2-lagged Total Sale
s as predictors
Furniture.MLR1 <- lm(Furniture~TotalSaleslag1 + TotalSaleslag2, data = total_Furniture.M
LR1)

# Multiple Regression Model 2
# Add 3-lagged Total Sales and remove missing values
total_Furniture$TotalSaleslag3 <- lag(total_Furniture$TotalSales, 3)
total_Furniture.MLR2 <- na.omit(total_Furniture)

# Build Furniture MLR2 predicting Furniture sales using 1-lagged, 2-lagged, and 3-lagged
Total Sales as predictors
Furniture.MLR2 <- lm(Furniture~TotalSaleslag1+TotalSaleslag2+TotalSaleslag3, data=total_
Furniture.MLR2)

# Multiple Regression Model 3
# Add CSI lag and remove missing values
total_Furniture$TotalSaleslag1 <- lag(total_Furniture$TotalSales, 1)
total_Furniture$CSIlag1 <- lag(total_Furniture$CSI, 1)
total_Furniture.MLR3 <- na.omit(total_Furniture)

# Build Furniture MLR3 predicting Furniture sales using 1-lagged Total Sales and CSI as
predictor
Furniture.MLR3 <- lm(Furniture ~ TotalSaleslag1+CSIlag1, data=total_Furniture.MLR3)
total_Furniture
```

Date <chr>	Furniture <dbl>	TotalSales <dbl>	CSI <dbl>	TotalSaleslag1 <dbl>	TotalSaleslag2 <dbl>	TotalSaleslag3 <dbl>	CSIlag <db
2015-01	8507.469	19546.163	98.1	NA	NA	NA	N
2015-02	1693.850	11678.994	95.4	19546.163	NA	NA	98
2015-03	3574.659	6716.044	93.0	11678.994	19546.163	NA	95

Date	Furniture	TotalSales	CSI	TotalSaleslag1	TotalSaleslag2	TotalSaleslag3	CSIla
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<db
2015-04	1803.958	12455.482	95.9	6716.044	11678.994	19546.163	93
2015-05	2798.821	15165.051	90.7	12455.482	6716.044	11678.994	95
2015-06	5544.520	11884.169	96.1	15165.051	12455.482	6716.044	90
2015-07	5157.028	10075.740	93.1	11884.169	15165.051	12455.482	96
2015-08	8061.612	26797.763	91.9	10075.740	11884.169	15165.051	93
2015-09	4496.787	17158.932	87.2	26797.763	10075.740	11884.169	91
2015-10	3565.225	10112.641	90.0	17158.932	26797.763	10075.740	87

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```
Furniture.SLR
```

```
##
## Call:
## lm(formula = Furniture ~ TotalSaleslag1, data = total_Furniture.SLR)
##
## Coefficients:
##    (Intercept)  TotalSaleslag1
##      3642.5794          0.1237
```

```
Furniture.MLR1
```

```
##
## Call:
## lm(formula = Furniture ~ TotalSaleslag1 + TotalSaleslag2, data = total_Furniture.MLR
1)
##
## Coefficients:
##    (Intercept)  TotalSaleslag1  TotalSaleslag2
##     4017.29977         0.14005        -0.03166
```

```
Furniture.MLR2
```

```
##
## Call:
## lm(formula = Furniture ~ TotalSaleslag1 + TotalSaleslag2 + TotalSaleslag3,
##     data = total_Furniture.MLR2)
##
## Coefficients:
##    (Intercept)  TotalSaleslag1  TotalSaleslag2  TotalSaleslag3
##     3446.37171         0.14237        -0.05621         0.05524
```

```
Furniture.MLR3
```

```
##
## Call:
## lm(formula = Furniture ~ TotalSaleslag1 + CSIlag1, data = total_Furniture.MLR3)
##
## Coefficients:
##    (Intercept)  TotalSaleslag1         CSIlag1
##      6335.7783          0.1277        -27.7962
```

Technology sales trend projection

```
# Create working dataframe with Technology, Total Sales, and CSI
total_Technology <- sqldf('select Date, Technology, `Total Sales`, CSI from monthly_cate
gory_sales
                          join total_monthly_sales using(Date)')
names(total_Technology)[3] <- "TotalSales"

# Create 1-lagged Total Sales values for Technology SLR
total_Technology$TotalSaleslag1 <- lag(total_Technology$TotalSales, 1)
total_Technology.SLR <- na.omit(total_Technology)

# Build Technology SLR predicting Technology sales using 1-lagged Total Sales as predict
or
Technology.SLR <- lm(Technology~TotalSaleslag1, data=total_Technology.SLR)

# Technology Multiple Regression Model 1
# Add 2-lagged Total Sales and remove missing values
total_Technology$TotalSaleslag2 <- lag(total_Technology$TotalSales, 2)
total_Technology.MLR1 <- na.omit(total_Technology)

# Build Technology MLR1 predicting Technology sales using 1-lagged and 2-lagged Total Sa
les as predictors
Technology.MLR1 <- lm(Technology~TotalSaleslag1+TotalSaleslag2, data=total_Technology.ML
R1)

# Multiple Regression Model 2
# Add 3-lagged Total Sales and remove missing values
total_Technology$TotalSaleslag3 <- lag(total_Technology$TotalSales, 3)
total_Technology.MLR2 <- na.omit(total_Technology)

# Build Technology MLR2 predicting Technology sales using 1-lagged, 2-lagged, and 3-lagg
ed Total Sales as predictors
Technology.MLR2 <- lm(Technology~TotalSaleslag1+TotalSaleslag2+TotalSaleslag3, data=tota
l_Technology.MLR2)

# Multiple Regression Model 3
# Add CSI lag and remove missing values
total_Technology$TotalSaleslag1 <- lag(total_Technology$TotalSales, 1)
total_Technology$CSIlag1 <- lag(total_Technology$CSI, 1)
total_Technology.MLR3 <- na.omit(total_Technology)

# Build Technology MLR3 predicting Technology sales using 1-lagged Total Sales and CSI a
s predictor
Technology.MLR3 <- lm(Technology ~ TotalSaleslag1+CSIlag1, data=total_Technology.MLR3)
total_Technology
```

Date	Technology	TotalSales	CSI	TotalSaleslag1	TotalSaleslag2	TotalSaleslag3	CSIl
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<(
2015-01	6436.330	19546.163	98.1	NA	NA	NA	
2015-02	5743.308	11678.994	95.4	19546.163	NA	NA	9

Date <chr>	Technology <dbl>	TotalSales <dbl>	CSI <dbl>	TotalSaleslag1 <dbl>	TotalSaleslag2 <dbl>	TotalSaleslag3 <dbl>	CSII <c
2015-03	1506.368	6716.044	93.0	11678.994	19546.163	NA	9
2015-04	6627.572	12455.482	95.9	6716.044	11678.994	19546.163	9
2015-05	2192.416	15165.051	90.7	12455.482	6716.044	11678.994	9
2015-06	2772.166	11884.169	96.1	15165.051	12455.482	6716.044	9
2015-07	2104.486	10075.740	93.1	11884.169	15165.051	12455.482	9
2015-08	13986.188	26797.763	91.9	10075.740	11884.169	15165.051	9
2015-09	6946.388	17158.932	87.2	26797.763	10075.740	11884.169	9
2015-10	3666.016	10112.641	90.0	17158.932	26797.763	10075.740	8

1-10 of 48 rows Previous **1** 2 3 4 5 Next

```
Technology.SLR
```

```
##
## Call:
## lm(formula = Technology ~ TotalSaleslag1, data = total_Technology.SLR)
##
## Coefficients:
##    (Intercept)  TotalSaleslag1
##      3708.2668          0.1789
```

```
Technology.MLR1
```

```
##
## Call:
## lm(formula = Technology ~ TotalSaleslag1 + TotalSaleslag2, data = total_Technology.ML
R1)
##
## Coefficients:
##    (Intercept)  TotalSaleslag1  TotalSaleslag2
##     4474.25218         0.21201        -0.07327
```

```
Technology.MLR2
```

```
##
## Call:
## lm(formula = Technology ~ TotalSaleslag1 + TotalSaleslag2 + TotalSaleslag3,
##     data = total_Technology.MLR2)
##
## Coefficients:
##    (Intercept)  TotalSaleslag1  TotalSaleslag2  TotalSaleslag3
##     4189.92514         0.20295        -0.08542         0.04169
```

```
Technology.MLR3
```

```
##
## Call:
## lm(formula = Technology ~ TotalSaleslag1 + CSIlag1, data = total_Technology.MLR3)
##
## Coefficients:
##    (Intercept)  TotalSaleslag1         CSIlag1
##     -3.269e+04       8.185e-02       4.039e+02
```

Office Supplies sales trend projection

```
# Create working dataframe with Office, Total Sales, and CSI
total_Office <- sqldf('select Date, Office, `Total Sales`, CSI from monthly_category_sal
es
                          join total_monthly_sales using(Date)')
names(total_Office)[3] <- "TotalSales"

# Create 1-lagged Total Sales values for Office SLR
total_Office$TotalSaleslag1 <- lag(total_Office$TotalSales, 1)
total_Office.SLR <- na.omit(total_Office)

# Build Office SLR predicting Office sales using 1-lagged Total Sales as predictor
Office.SLR <- lm(Office~TotalSaleslag1, data=total_Office.SLR)

# Office Multiple Regression Model 1
# Add 2-lagged Total Sales and remove missing values
total_Office$TotalSaleslag2 <- lag(total_Office$TotalSales, 2)
total_Office.MLR1 <- na.omit(total_Office)

# Build Office MLR1 predicting Office sales using 1-lagged and 2-lagged Total Sales as p
redictors
Office.MLR1 <- lm(Office~TotalSaleslag1+TotalSaleslag2, data=total_Office.MLR1)

# Multiple Regression Model 2
# Add 3-lagged Total Sales and remove missing values
total_Office$TotalSaleslag3 <- lag(total_Office$TotalSales, 3)
total_Office.MLR2 <- na.omit(total_Office)

# Build Office MLR2 predicting Office sales using 1-lagged, 2-lagged, and 3-lagged Total
Sales as predictors
Office.MLR2 <- lm(Office~TotalSaleslag1+TotalSaleslag2+TotalSaleslag3, data=total_Offic
e.MLR2)

# Multiple Regression Model 3
# Add CSI lag and remove missing values
total_Office$TotalSaleslag1 <- lag(total_Office$TotalSales, 1)
total_Office$CSIlag1 <- lag(total_Office$CSI, 1)
total_Office.MLR3 <- na.omit(total_Office)

# Build Office MLR3 predicting Office sales using 1-lagged Total Sales and CSI as predic
tor
Office.MLR3 <- lm(Office ~ TotalSaleslag1+CSIlag1, data=total_Office.MLR3)
total_Office
```

Date <chr>	Office <dbl>	TotalSales <dbl>	CSI <dbl>	TotalSaleslag1 <dbl>	TotalSaleslag2 <dbl>	TotalSaleslag3 <dbl>	CSIlag <db
2015-01	4602.364	19546.163	98.1	NA	NA	NA	N
2015-02	4241.836	11678.994	95.4	19546.163	NA	NA	98
2015-03	1635.017	6716.044	93.0	11678.994	19546.163	NA	95

Date	Office	TotalSales	CSI	TotalSaleslag1	TotalSaleslag2	TotalSaleslag3	CSIla
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<db
2015-04	4023.952	12455.482	95.9	6716.044	11678.994	19546.163	93
2015-05	10173.814	15165.051	90.7	12455.482	6716.044	11678.994	95
2015-06	3567.483	11884.169	96.1	15165.051	12455.482	6716.044	90
2015-07	2814.226	10075.740	93.1	11884.169	15165.051	12455.482	96
2015-08	4749.963	26797.763	91.9	10075.740	11884.169	15165.051	93
2015-09	5715.757	17158.932	87.2	26797.763	10075.740	11884.169	91
2015-10	2881.400	10112.641	90.0	17158.932	26797.763	10075.740	87

1-10 of 48 rows Previous **1** 2 3 4 5 Next

```
Office.SLR
```

```
##
## Call:
## lm(formula = Office ~ TotalSaleslag1, data = total_Office.SLR)
##
## Coefficients:
##    (Intercept)  TotalSaleslag1
##      2774.2559          0.1379
```

```
Office.MLR1
```

```
##
## Call:
## lm(formula = Office ~ TotalSaleslag1 + TotalSaleslag2, data = total_Office.MLR1)
##
## Coefficients:
##    (Intercept)  TotalSaleslag1  TotalSaleslag2
##     3109.39185         0.15235        -0.03139
```

```
Office.MLR2
```

```
##
## Call:
## lm(formula = Office ~ TotalSaleslag1 + TotalSaleslag2 + TotalSaleslag3,
##     data = total_Office.MLR2)
##
## Coefficients:
##    (Intercept)  TotalSaleslag1  TotalSaleslag2  TotalSaleslag3
##      3.179e+03       1.433e-01      -2.831e-02       5.408e-03
```

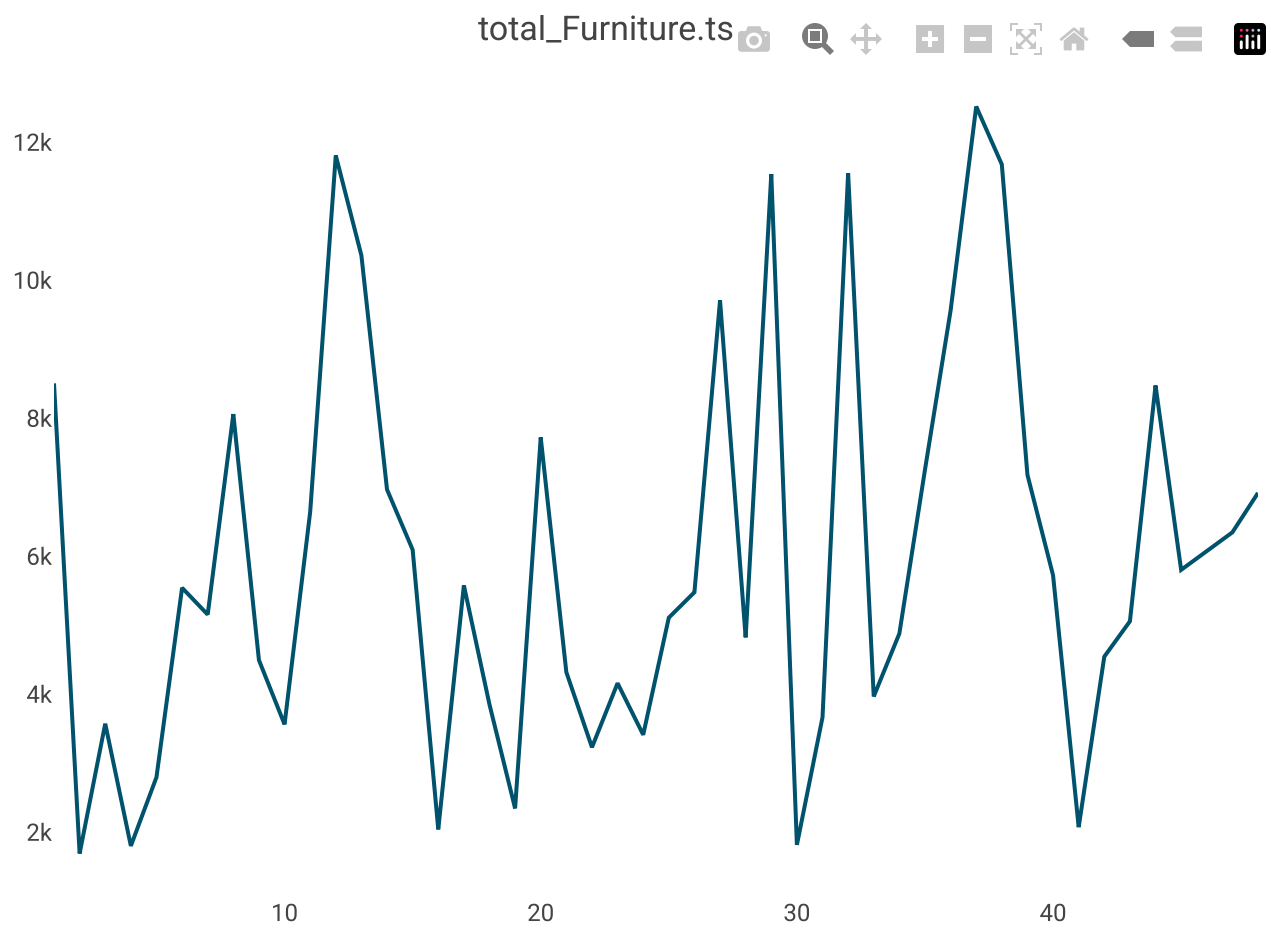
```
Office.MLR3
```

```
##
## Call:
## lm(formula = Office ~ TotalSaleslag1 + CSIlag1, data = total_Office.MLR3)
##
## Coefficients:
##    (Intercept)  TotalSaleslag1         CSIlag1
##     -1.519e+04       8.835e-02       1.999e+02
```

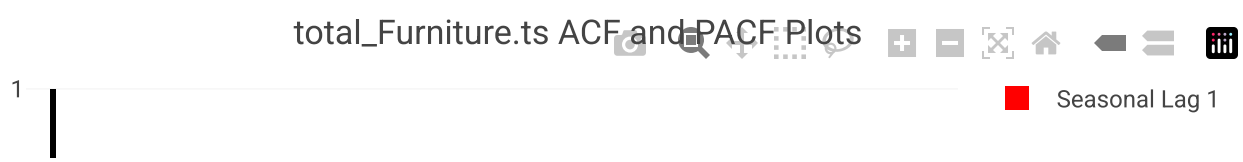
ARIMA Models:

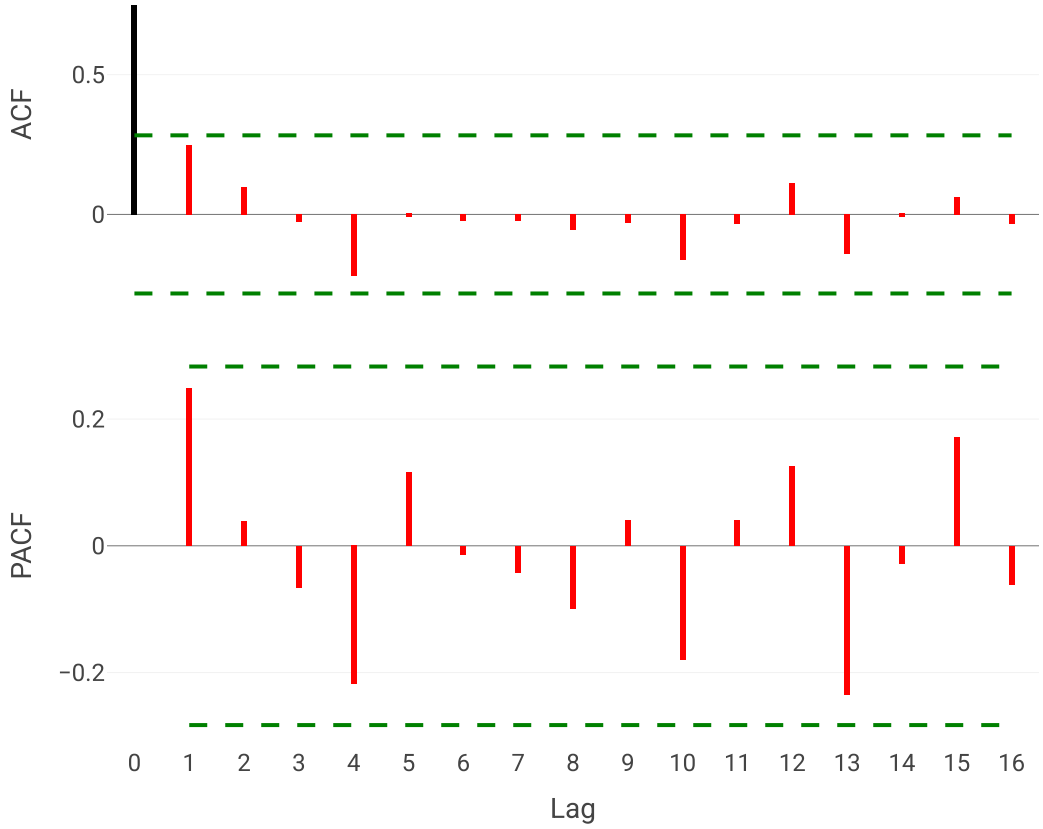
Furniture sales ARIMA model

```
total_Furniture.ts <- ts(monthly_category_sales$Furniture)
ts_plot(total_Furniture.ts)
```




```
ts_cor(total_Furniture.ts)
```






```
# use BIC to identify the top three candidate models
auto.arima(total_Furniture.ts[1:33], trace=TRUE, stationary=FALSE, seasonal=FALSE, ic=c
("bic"))
```

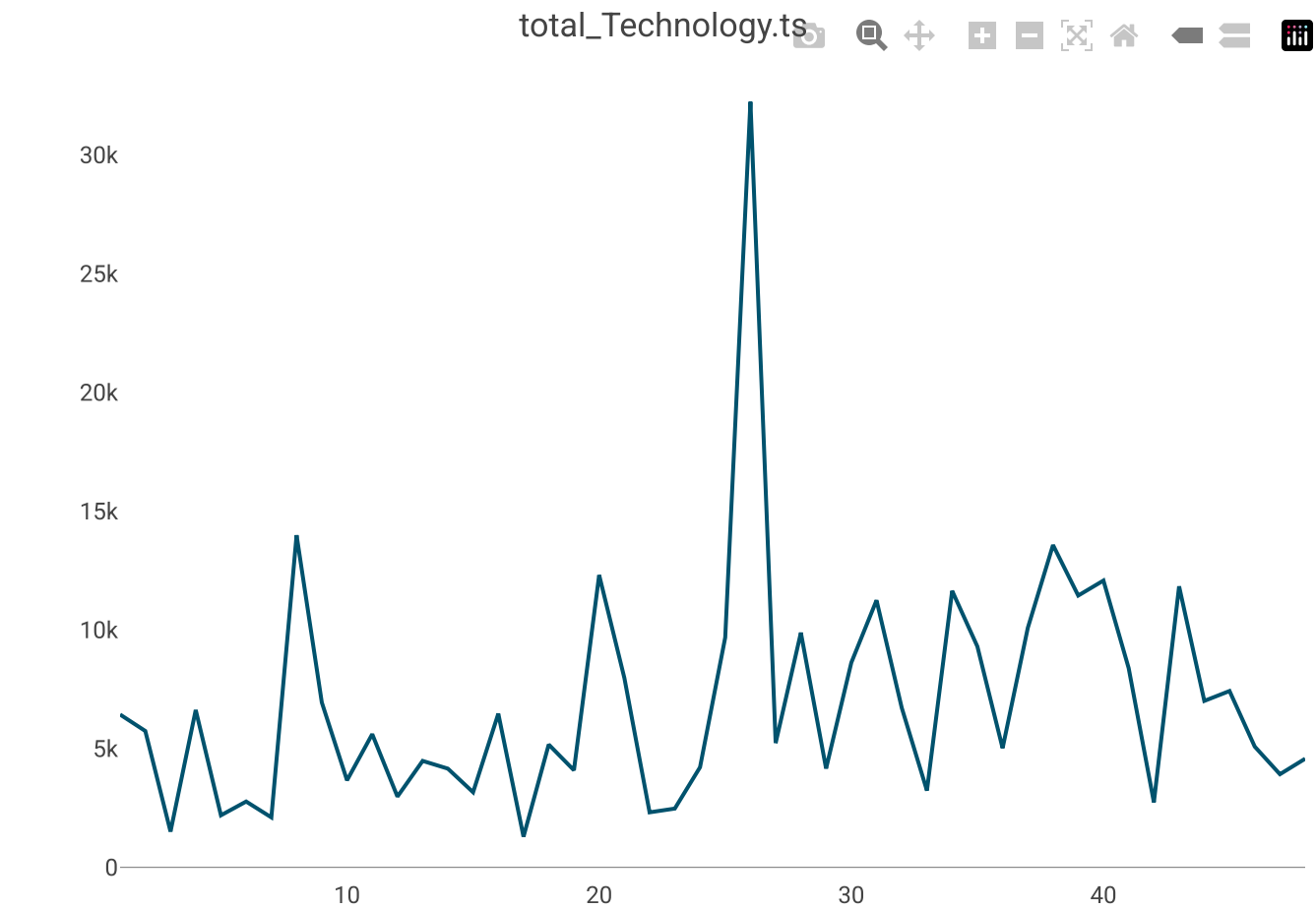
```
##
##  ARIMA(2,0,2) with non-zero mean : Inf
##  ARIMA(0,0,0) with non-zero mean : 627.0999
##  ARIMA(1,0,0) with non-zero mean : 630.5054
##  ARIMA(0,0,1) with non-zero mean : 630.5116
##  ARIMA(0,0,0) with zero mean     : 673.7109
##  ARIMA(1,0,1) with non-zero mean : 632.586
##
##  Best model: ARIMA(0,0,0) with non-zero mean
```

```
## Series: total_Furniture.ts[1:33]
## ARIMA(0,0,0) with non-zero mean
##
## Coefficients:
##            mean
##       5498.3162
## s.e.   506.9274
##
## sigma^2 = 8745166:  log likelihood = -310.05
## AIC=624.11   AICc=624.51   BIC=627.1
```

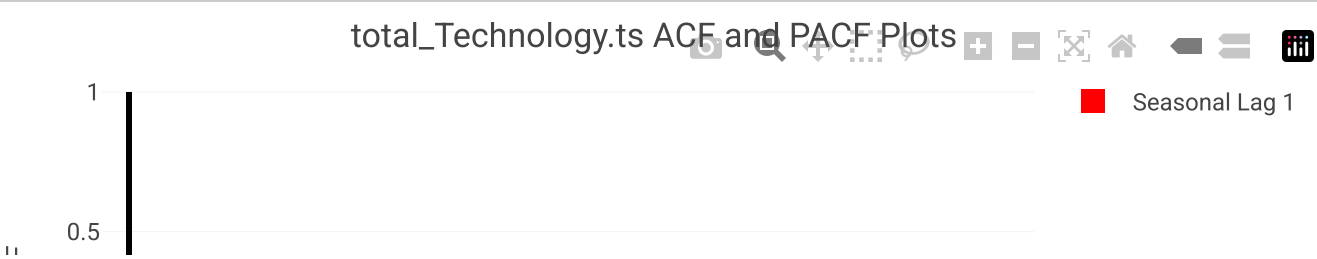
```
# build the top three candidate models using the training data
Furniture.candidate1<-Arima(total_Furniture.ts[1:36], order=c(1,0,1))
Furniture.candidate2<-Arima(total_Furniture.ts[1:36], order=c(2,0,0))
Furniture.candidate3<-Arima(total_Furniture.ts[1:36], order=c(1,0,0))

# apply the top three candidate models to predict for the most recent validation data ob
servations
Furniture.validation1<-Arima(total_Furniture.ts[37:48], model=Furniture.candidate1)
Furniture.validation2<-Arima(total_Furniture.ts[37:48], model=Furniture.candidate2)
Furniture.validation3<-Arima(total_Furniture.ts[37:48], model=Furniture.candidate3)
```

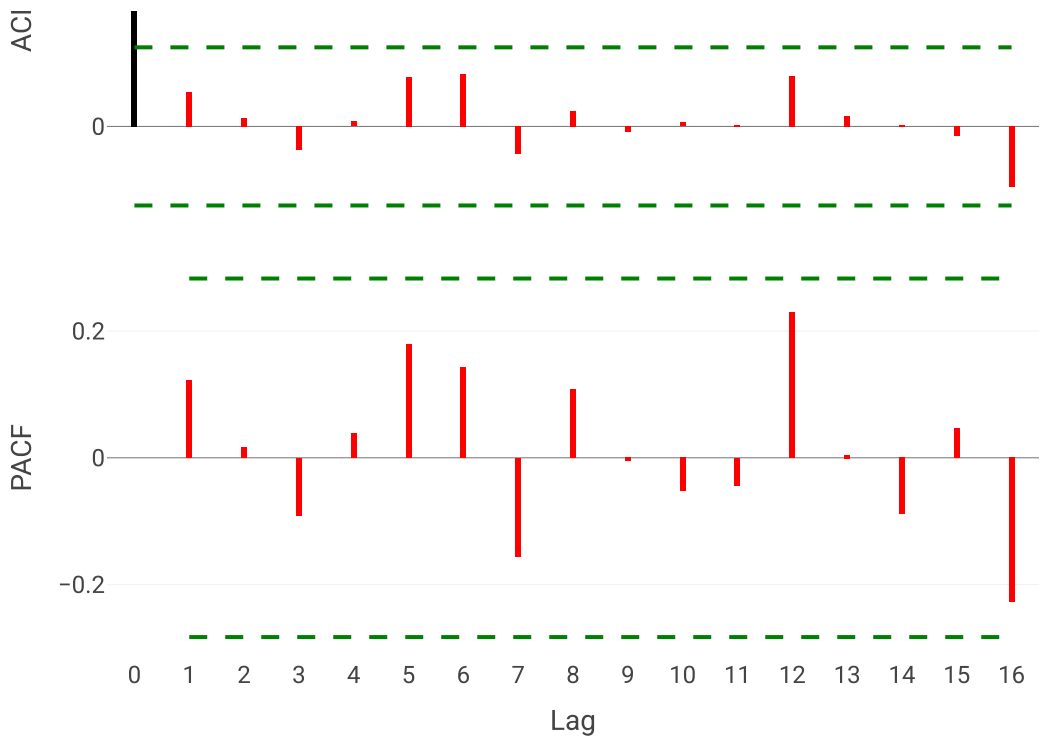
Technology sales ARIMA model

```
total_Technology.ts <- ts(monthly_category_sales$Technology)
ts_plot(total_Technology.ts)
```




```
ts_cor(total_Technology.ts)
```






```
# use BIC to identify the top three candidate models
auto.arima(total_Technology.ts[1:33], trace=TRUE, stationary=FALSE, seasonal=FALSE, ic=c
("bic"))
```

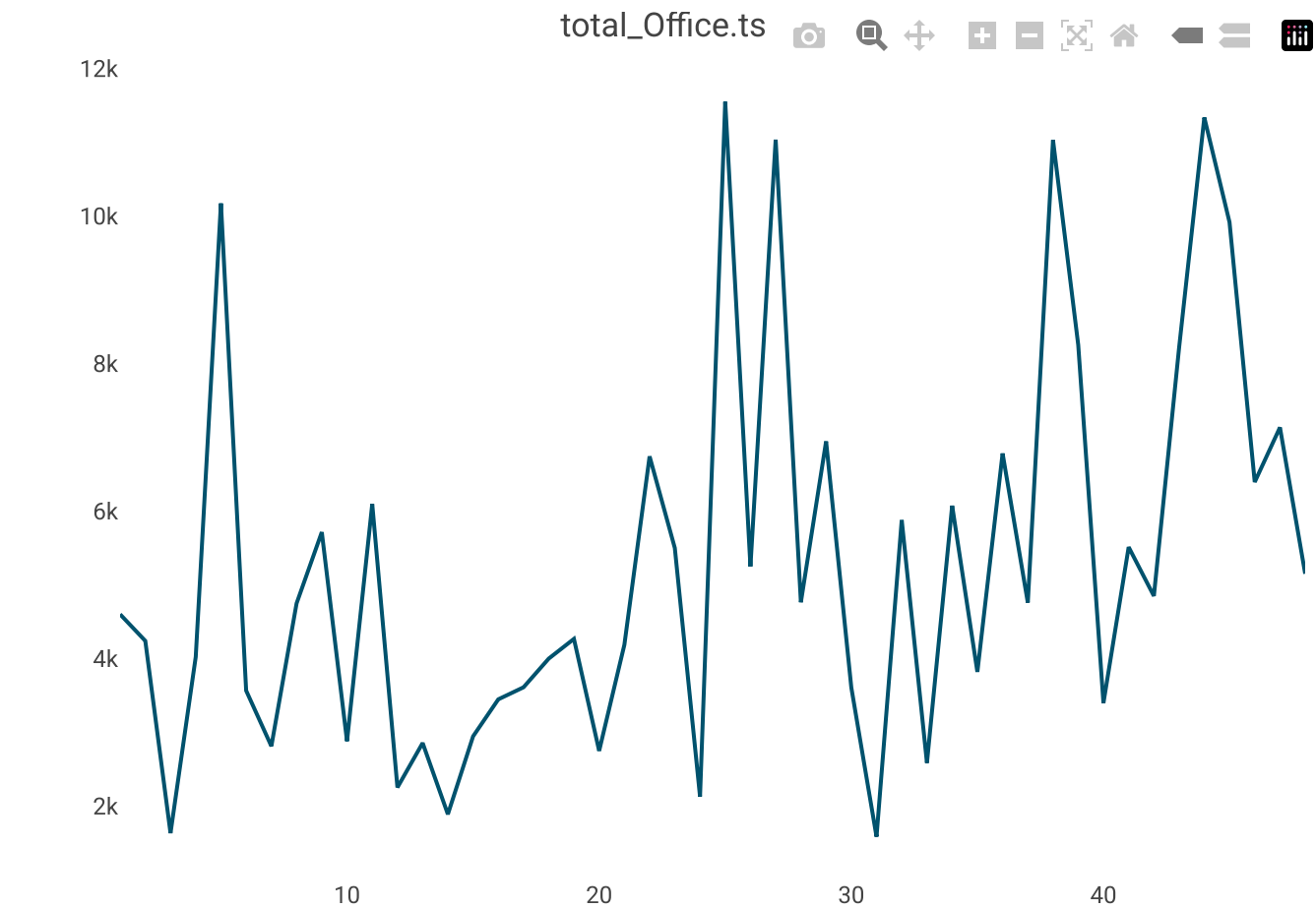
```
##
##  ARIMA(2,0,2) with non-zero mean : Inf
##  ARIMA(0,0,0) with non-zero mean : 669.5534
##  ARIMA(1,0,0) with non-zero mean : 672.797
##  ARIMA(0,0,1) with non-zero mean : 672.8063
##  ARIMA(0,0,0) with zero mean     : 693.7652
##  ARIMA(1,0,1) with non-zero mean : 676.2935
##
##  Best model: ARIMA(0,0,0) with non-zero mean
```

```
## Series: total_Technology.ts[1:33]
## ARIMA(0,0,0) with non-zero mean
##
## Coefficients:
##            mean
##       6354.8935
## s.e.   964.4937
##
## sigma^2 = 31657407:  log likelihood = -331.28
## AIC=666.56   AICc=666.96   BIC=669.55
```

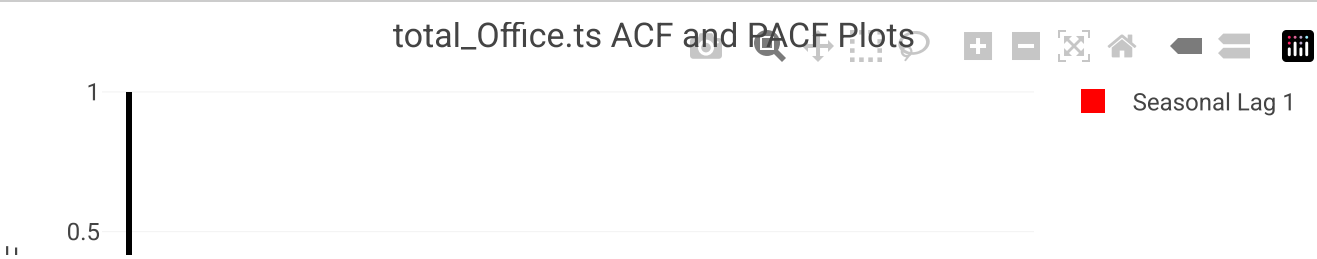
```
# build the top three candidate models using the training data
Technology.candidate1<-Arima(total_Technology.ts[1:36], order=c(1,0,1))
Technology.candidate2<-Arima(total_Technology.ts[1:36], order=c(2,0,0))
Technology.candidate3<-Arima(total_Technology.ts[1:36], order=c(1,0,0))

# apply the top three candidate models to predict for the most recent validation data ob
servations
Technology.validation1<-Arima(total_Technology.ts[37:48], model=Technology.candidate1)
Technology.validation2<-Arima(total_Technology.ts[37:48], model=Technology.candidate2)
Technology.validation3<-Arima(total_Technology.ts[37:48], model=Technology.candidate3)
```

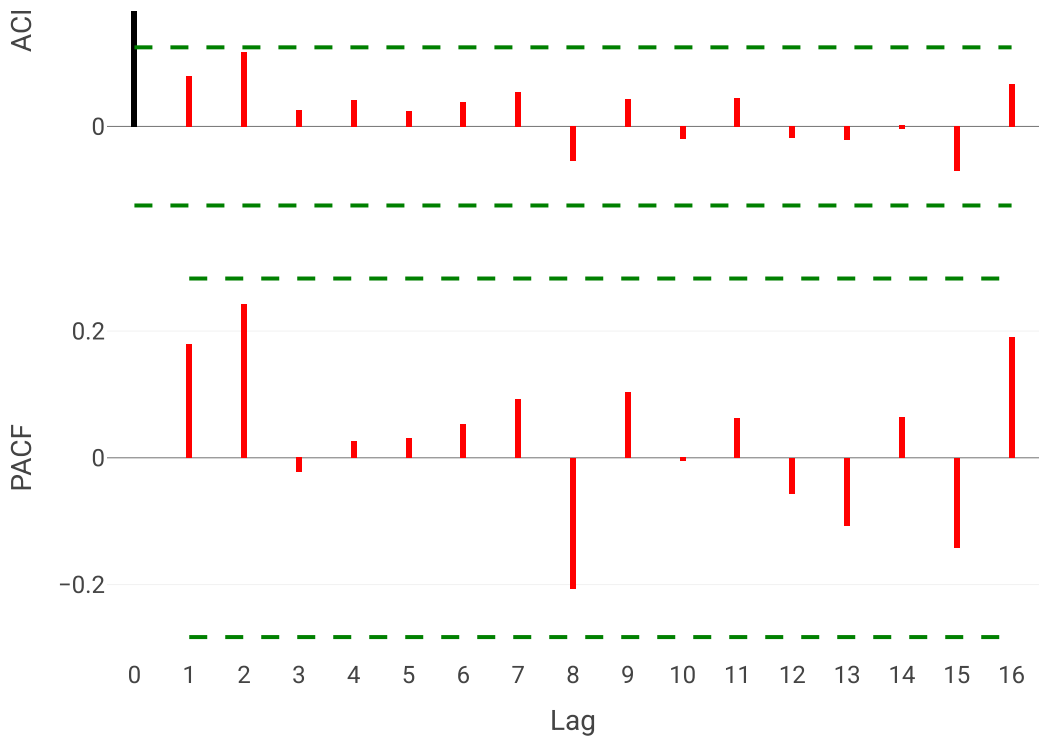
Office Supplies sales ARIMA model

```
total_Office.ts <- ts(monthly_category_sales$Office)
ts_plot(total_Office.ts)
```




```
ts_cor(total_Office.ts)
```






```
# use BIC to identify the top three candidate models
auto.arima(total_Office.ts[1:33], trace=TRUE, stationary=FALSE, seasonal=FALSE, ic=c("bi
c"))
```

```
##
##  ARIMA(2,0,2) with non-zero mean : Inf
##  ARIMA(0,0,0) with non-zero mean : 615.8309
##  ARIMA(1,0,0) with non-zero mean : 619.2759
##  ARIMA(0,0,1) with non-zero mean : 619.2916
##  ARIMA(0,0,0) with zero mean     : 661.5434
##  ARIMA(1,0,1) with non-zero mean : 622.3485
##
##  Best model: ARIMA(0,0,0) with non-zero mean
```

```
## Series: total_Office.ts[1:33]
## ARIMA(0,0,0) with non-zero mean
##
## Coefficients:
##            mean
##       4554.8775
## s.e.   427.3595
##
## sigma^2 = 6215317:  log likelihood = -304.42
## AIC=612.84   AICc=613.24   BIC=615.83
```

```
# build the top three candidate models using the training data
Office.candidate1<-Arima(total_Office.ts[1:36], order=c(1,0,1))
Office.candidate2<-Arima(total_Office.ts[1:36], order=c(2,0,0))
Office.candidate3<-Arima(total_Office.ts[1:36], order=c(1,0,0))

# apply the top three candidate models to predict for the most recent validation data ob
servations
Office.validation1<-Arima(total_Office.ts[37:48], model=Office.candidate1)
Office.validation2<-Arima(total_Office.ts[37:48], model=Office.candidate2)
Office.validation3<-Arima(total_Office.ts[37:48], model=Office.candidate3)
```

b. Model performance comparison

Furniture sales trend projection model Adjusted-Rsq values

```
# report the Adjusted-Rsq values using "summary()$adj.r.squared" command
summary(Furniture.SLR)$adj.r.squared
```

```
## [1] 0.07525267
```

```
summary(Furniture.MLR1)$adj.r.squared
```

```
## [1] 0.06755171
```

```
summary(Furniture.MLR2)$adj.r.squared
```

```
## [1] 0.05219964
```

```
summary(Furniture.MLR3)$adj.r.squared
```

```
## [1] 0.05453948
```

Technology sales trend projection model Adjusted-Rsq values

```
# report the Adjusted-Rsq values using "summary()$adj.r.squared" command
summary(Technology.SLR)$adj.r.squared
```

```
## [1] 0.04452869
```

```
summary(Technology.MLR1)$adj.r.squared
```

```
## [1] 0.03172554
```

```
summary(Technology.MLR2)$adj.r.squared
```

```
## [1] -0.0003237456
```

```
summary(Technology.MLR3)$adj.r.squared
```

```
## [1] 0.08144633
```

Office Supplies sales trend projection model Adjusted-Rsq values

```
# report the Adjusted-Rsq values using "summary()$adj.r.squared" command
summary(Office.SLR)$adj.r.squared
```

```
## [1] 0.1204249
```

```
summary(Office.MLR1)$adj.r.squared
```

```
## [1] 0.107459
```

```
summary(Office.MLR2)$adj.r.squared
```

```
## [1] 0.0701949
```

```
summary(Office.MLR3)$adj.r.squared
```

```
## [1] 0.149248
```

Decide the best trend projection model for each sales category:

The best trend projection model for Furniture sales is Furniture.SLR

The best trend projection model for Technology sales is Technology.MLR3

The best trend projection model for Office Supplies sales is Office.MLR3

Furniture sales ARIMA candidate model MAPE

```
# calculate MAPE for each of the top three candidate models
accuracy(Furniture.validation1)
```

```
##                    ME     RMSE      MAE        MPE     MAPE     MASE      ACF1
## Training set 1138.28 2876.413 2044.781 -0.7511075 33.35877 1.090103 0.4089465
```

```
accuracy(Furniture.validation2)
```

```
##                     ME     RMSE      MAE      MPE     MAPE     MASE      ACF1
## Training set 1118.001 2917.818 1963.745 -1.65394 32.34373 1.046901 0.4472614
```

```
accuracy(Furniture.validation3)
```

```
##                     ME     RMSE      MAE       MPE     MAPE     MASE      ACF1
## Training set 1122.396 2918.028 1965.457 -1.569979 32.35095 1.047814 0.4472253
```

Technology sales ARIMA candidate model MAPE

```
# calculate MAPE for each of the top three candidate models
accuracy(Technology.validation1)
```

```
##                     ME     RMSE      MAE       MPE     MAPE     MASE      ACF1
## Training set 1503.284 3710.852 3124.718 -2.788331 43.19449 1.008806 0.2631366
```

```
accuracy(Technology.validation2)
```

```
##                     ME     RMSE      MAE       MPE     MAPE     MASE      ACF1
## Training set 1521.895 3721.457 3134.218 -2.489908 43.16191 1.011873 0.2626875
```

```
accuracy(Technology.validation3)
```

```
##                     ME     RMSE      MAE       MPE    MAPE     MASE      ACF1
## Training set 1498.231 3708.007 3122.187 -2.870463 43.2057 1.007989 0.2633958
```

Office Supplies sales ARIMA candidate model MAPE

```
# calculate MAPE for each of the top three candidate models
accuracy(Office.validation1)
```

```
##                     ME     RMSE      MAE     MPE    MAPE     MASE      ACF1
## Training set 2726.108 3776.657 2933.094 28.9315 35.0205 1.046378 0.1988166
```

```
accuracy(Office.validation2)
```

```
##                    ME     RMSE      MAE      MPE     MAPE      MASE      ACF1
## Training set 2102.34 3462.998 2534.738 18.47203 31.11716 0.9042648 0.2105828
```

```
accuracy(Office.validation3)
```

```
##                    ME     RMSE      MAE      MPE     MAPE    MASE      ACF1
## Training set 2693.082 3704.941 2859.184 28.83744 33.72375 1.02001 0.1935728
```

The best ARIMA model for Furniture sales is ARIMA(2,0,0) with non-zero mean.

The best ARIMA model for Technology sales is ARIMA(2,0,0) with non-zero mean.

The best ARIMA model for Office Supplies is ARIMA(2,0,0) with non-zero mean.

c. Data forecasting

Forecast for Furniture sales for next month

```
# 1. use the best trend projection model + AR(1) model for Furniture sales to produce th
e upper bound of 90% range forecast for next week.

# peel off the predicted trend from the observed Furniture sales to obtain their residua
ls.
total_Furniture.SLR$Residual<-total_Furniture.SLR$Furniture-Furniture.SLR$fitted.values

# use the ts_cor() to create the ACF and PACF charts of the residual data
Residual.ts<-ts(total_Furniture.SLR$Residual)

# Use AR(1) Model to predict the Furniture sales residual data.
Furniture.AR.model <- Arima(Residual.ts, order = c(1,0,0))
# Add the epsilon column (i.e., after peeling off trend and autocorrelation components)
# into the dataframe "CSI.MLR3$epsilon"
total_Furniture.SLR$epsilon<-Furniture.AR.model$residuals

# produce forecasts of Furniture sales residual for the next month
Furniture.residual.forecast<- forecast(Furniture.AR.model, h = 1, level=c(90)) # h=1 ind
icates one time step forward forecast; level=c(90) indicates we build 90% range forecast

# produce forecasts of trend of Furniture sales in next month
Furniture.newdata<-data.frame(TotalSaleslag1=total_Furniture$TotalSaleslag1[48])
Furniture.trend.forecast<- predict(Furniture.SLR, # using the best trend model "SLR" to
make the forecast
        newdata = Furniture.newdata, # provide the predictor value
        interval = "confidence", # output the range forecast for the trend
        level=0.90 # with 90% probability
)

Furniture.trend.forecast<-data.frame(Furniture.trend.forecast)  # convert "trend.forecas
t" into a data frame

# produce a forecast value for Furniture sales in next month
Furniture.sales.forecast1<-Furniture.trend.forecast$upr + as.numeric(Furniture.residual.
forecast$upper) # upper bound of 90% range forecast of Furniture sales in next month (No
te: the result is not really mathematically correct, but good enough for practical purpo
se)
Furniture.sales.forecast1
```

```
## [1] 11253.31
```

```
# 2. use the best ARIMA model for Furniture to produce the upper bound of 90% range fore
cast for next month
Furniture.fit.best<-Arima(total_Furniture.ts[1:48], order = c(2, 0, 0))
Furniture_sales.nextweek<- forecast(Furniture.fit.best, h = 1, level=c(90))
Furniture.sales.forecast2<-as.numeric(Furniture_sales.nextweek$upper)
Furniture.sales.forecast2
```

```
## [1] 10987.47
```

Forecast for Technology sales for next month

```
# 1. use the best trend projection model + AR(1) model for Technology sales to produce t
he upper bound of 90% range forecast for next week.

# peel off the predicted trend from the observed Technology sales to obtain their residu
als.
total_Technology.MLR3$Residual<-total_Technology.MLR3$Technology-Technology.MLR3$fitted.
values

# use the ts_cor() to create the ACF and PACF charts of the residual data
Residual.ts<-ts(total_Technology.MLR3$Residual)

# Use AR(1) Model to predict the Technology sales residual data.
Technology.AR.model <- Arima(Residual.ts, order = c(1,0,0))
# Add the epsilon column (i.e., after peeling off trend and autocorrelation components)
# into the dataframe "CSI.MLR3$epsilon"
total_Technology.MLR3$epsilon<-Technology.AR.model$residuals

# produce forecasts of Technology sales residual for the next month
Technology.residual.forecast<- forecast(Technology.AR.model, h=1, level=c(90)) # h=1 ind
icates one time step forward forecast; level=c(90) indicates we build 90% range forecast

# produce forecasts of trend of Technology sales in next month
Technology.newdata<-data.frame(TotalSaleslag1=total_Technology$TotalSaleslag1[48],
                               CSIlag1=total_Technology$CSI[48])
Technology.trend.forecast<- predict(Technology.MLR3, # using the best trend model "MLR3"
to make the forecast
        newdata = Technology.newdata, # provide the predictor value
        interval = "confidence", # output the range forecast for the trend
        level=0.90 # with 90% probability
)

Technology.trend.forecast<-data.frame(Technology.trend.forecast) # convert "trend.foreca
st" into a data frame

# produce a forecast value for Technology sales in next month
Technology.sales.forecast1<-Technology.trend.forecast$upr + as.numeric(Technology.residu
al.forecast$upper) # upper bound of 90% range forecast of Technology sales in next month
(Note: the result is not really mathematically correct, but good enough for practical pu
rpose)
Technology.sales.forecast1
```

```
## [1] 18689.15
```

```
# 2. use the best ARIMA model for Technology to produce the upper bound of 90% range for
ecast for next month
Technology.fit.best<-Arima(total_Technology.ts[1:48], order = c(2, 0, 0))
Technology_sales.nextweek<- forecast(Technology.fit.best, h = 1, level=c(90))
Technology.sales.forecast2<-as.numeric(Technology_sales.nextweek$upper)
Technology.sales.forecast2
```

```
## [1] 15105.9
```

Forecast for Office Supplies sales for next month

```
# 1. use the best trend projection model + AR(1) model for Office sales to produce the u
pper bound of 90% range forecast for next week.

# peel off the predicted trend from the observed Office sales to obtain their residuals.
total_Office.MLR3$Residual<-total_Office.MLR3$Office-Office.MLR3$fitted.values

# use the ts_cor() to create the ACF and PACF charts of the residual data
Residual.ts<-ts(total_Office.MLR3$Residual)

# Use AR(1) Model to predict the Office sales residual data.
Office.AR.model <- Arima(Residual.ts, order = c(1,0,0))
# Add the epsilon column (i.e., after peeling off trend and autocorrelation components)
# into the dataframe "CSI.MLR3$epsilon"
total_Office.MLR3$epsilon<-Office.AR.model$residuals

# produce forecasts of Office sales residual for the next month
Office.residual.forecast<- forecast(Office.AR.model, h = 1, level=c(90)) # h=1 indicates
one time step forward forecast; level=c(90) indicates we build 90% range forecast

# produce forecasts of trend of Office sales in next month
Office.newdata<-data.frame(TotalSaleslag1=total_Office$TotalSaleslag1[48],
                           CSIlag1=total_Office$CSI[48])
Office.trend.forecast<- predict(Office.MLR3, # using the best trend model "MLR3" to make
the forecast
        newdata = Office.newdata, # provide the predictor value
        interval = "confidence", # output the range forecast for the trend
        level=0.90 # with 90% probability
)

Office.trend.forecast<-data.frame(Office.trend.forecast) # convert "trend.forecast" into
a data frame

# produce a forecast value for Office sales in next month
Office.sales.forecast1<-Office.trend.forecast$upr  +  as.numeric(Office.residual.forecast
$upper) # upper bound of 90% range forecast of Office sales in next month (Note: the res
ult is not really mathematically correct, but good enough for practical purpose)
Office.sales.forecast1
```

```
## [1] 11060.74
```

```
# 2. use the best ARIMA model for Office to produce the upper bound of 90% range forecas
t for next month
Office.fit.best<-Arima(total_Office.ts[1:48], order = c(2, 0, 0))
Office_sales.nextweek<- forecast(Office.fit.best, h = 1, level=c(90))
Office.sales.forecast2<-as.numeric(Office_sales.nextweek$upper)
Office.sales.forecast2
```

```
## [1] 9998.669
```

4. Discussion and Recommendation

Best Regression + AR(1) Model: Office Supplies MLR3 Adjusted-Rsq: 0.149248 Forecast: 11060.74

Best ARIMA Model: Office Supplies ARIMA(2,0,0)
MAPE: 31.11716 Forecast: 9685.941 ((2,0,0) was chosen because when considering MAPE as well as BIC, it is better than just BIC alone using (0,0,0) which was not meaningfully different than other ARIMA orders.

Initially in this analysis, two other predictor factors were used in the MLR3 models. The first, a metric called Consumer Confidence Index which measures consumers' level of financial comfortability based on current income, costs and short-term future purchasing intentions such as vactions, cars, houses and other discretionary spending. However, this metric did show enough variance over the relevant dates range of Jan 2015-Dec 2018, and did not result in meaningful Regression + AR(1) Adjusted-Rsq or ARIMA models. The Adjusted-Rsq values were all extremely low, or negative in the case of the Technology category. Another predictor variable was used - the 30 Year Mortgage Rate. This showed minor improvements but still did not result in helpful or meaningful model results. The only case where model results showed positive correlation were in the Office Supplies MLR3, however this was likely due to spurious correlation because if mortgage rates and office supplies really are correlated it would likely show in the other categories as well.

Ultimately, a third factor was chosen - Consumer Sentiment Index. This index is similar to Consumer Confidence Index, however the survey in which the data is collected take into account more specific purchasing patterns such as household goods, and whether the survey responder feels more financially comfortable than one year ago, and their sentiment on the upcoming year. This metric showed somewhat more variance than the Consumer Confidence Index, and is more stationary data than the 30-Year Mortgage rate. This resulted in much better model results, and showed meaningful correlation 2 out of 3 categories, Technology and Office Supplies. Only the Furniture category SLR model was marginally better than the Furniture MLR3 model.

Forecasting results were all within reasonable bounds based on historical data, and only the Technology sales forecast shows a somewhat higher prediction. This would make sense due to Technology sales historically showing spikes in months December and January.

The recommendation from these findings is that using recent sales performance within only 1-2 months is more useful than 3 or more, and that the Consumer Sentiment Index is a better outside predicting factor to use for multimple regression models. Because these categories showed positive correlation, it can be assumed that total store sales across all categories will increase with the Consumer Sentiment Index.